Sihong Yu

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Open

WORK EXPERIENCE

Full Stack Developer (Tech Lead) **ReCube, Hong Kong** **Next.js, AWS** **Apr 2023 - Feb 2024**

- Technical lead; led a 3-person team to deliver 80+ features across consumer and restaurant apps, pass UAT, and launch.
- Built a Next.js SSR PWA with AWS CDK/Amplify CI/CD pipelines to accelerate iteration.
- Led the container rental platform from scratch, using AWS DynamoDB for unique utensil IDs and borrow/return tracking.
- Integrated Stripe card payments with deposit pre-authorization and refunds, applying behavioral design to boost return compliance.
- Integrated Firebase Auth email/SMS for user authentication and access control.
- Developed an AWS EventBridge/SES notification system to push orders and reminders, driving on-time borrow/return.
- Managed Cloudflare domains and routing for development, testing, and production environments.
- Designed a reusable i18n hook enabling one-click English/Traditional Chinese switching, decoupling content from code.
- Produced and deployed in-app tutorial videos to lower onboarding friction and improve retention.

PROJECT

BioVis **Mixed Reality Biosensing • Research Project** **Unity, Meta Quest 3, C#** **May 2025 – Nov 2025**

- Co-authored (2nd) a Siggraph Asia XR submission; built the MR biosensing prototype.
- Solo-built a Unity project with 6 scenes and 40+ scripts, integrating 3D assets and UI.
- Used ComputeShader for parallel image processing; reproduced EVM; deployed a Quest 3 real-time MR prototype.
- Evaluated ONNX and Sentis quantization/inference; benchmarked YOLO v11 nano in real time on Quest 3.
- Reproduced multiple rPPG algorithms and validated across datasets, delivering academic-to-edge real-time deployment.

ShYu Resume **Personal Resume • Open Source Project** **Next.js, Tailwind** **Aug 2024 – Present**

- Built a Next.js i18n resume generator with tri-language (ZH/EN/HK) switching and A4 PDF export (hyperlinks included).
- Deployed to GitHub Pages with one-click save/print.

Carbon 2 Garden **Sensor Interactive App • COMP90018** **React Native, Firebase** **Sep 2024 – Nov 2024**

- Built a React Native eco-habit app with tasks, leaderboards, and activity reminders.
- Integrated gyroscope and Google Maps API to ship a "sedentary alert + park recommendation" challenge prototype.
- Owned routing and project skeleton; shipped the APK.

3D Object Detection **Point Cloud • Bachelor Thesis** **Python, PyTorch, Open3D** **Sep 2022 – May 2023**

- Reproduced 3D object detection on OpenPCDet, generating vehicle/pedestrian/cyclist boxes on KITTI.
- Trained on a SLURM cluster; used Open3D for point-cloud visualization and experiment analysis.
- Completed the bachelor thesis independently, consolidating methods and results.

Raft Consensus Implementation in Go **Distributed Systems • CSCI4160** **Go, Raft** **Jan 2022 – May 2022**

- Implemented Raft per the paper; passed leader election and log replication tests.
- Delivered a linearizable distributed KV store; fixed lock contention and log commit issues; passed all tests.

Data & Algorithm Delivery **Course Project Collection** **SQL, Hadoop, Python** **Sep 2021 – May 2024**

- Designed Hadoop/MapReduce pipelines for parallel Dijkstra, PageRank, K-Means; optimized big-data processing flow.
- Designed a 5-table PK/FK schema and 8 complex SQL queries (multi-join/subquery/custom formulas) for league/team/sponsor analytics.
- Built PostGIS spatial SQL for distance, routing, single-point elevation, and area slope statistics.
- Splendor planning agents modeled with MDP and reward functions; tuned to beat the TA baseline (27–13).

EDUCATION

MSc. Computer Science (with Distinction) **The University of Melbourne**
WAM: 81.938/100 **Feb 2024 - Dec 2025**

BSc. Computer Science **The Chinese University of Hong Kong**
MGPA: 3.364/4.0 **Sep 2019 - Jul 2023**

- Honours at Entrance
- Dean's List, 2022-2023
- ELITE Stream

SKILLS

AI / Computer Vision	PyTorch · Open3D · 3D Object Detection · EVM/rPPG · Unity (C#, ComputeShader)
Cloud Computing & Infrastructure	AWS (CDK/Lambda/EventBridge/Amplify) · Firebase · GitHub Actions
Software Engineering	Next.js · TypeScript · React Native
Languages	English · Mandarin · Cantonese